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Inventor(s)	Yuan; Zheng et al.

Streaming secondary device content on a web conference

Abstract

In an embodiment, method and system includes initiating a web conferencing meeting, detecting a plurality of client devices for to participate in the web conferencing meeting, detecting a local source associated with one of the plurality of client devices, selecting from content from a local source to share with other client devices during the web conferencing meeting, and streaming the content from the local source through one of the plurality of client devices associated with the local source.

Inventors:	Yuan; Zheng (San Jose, CA), Sheng; Hailei (San Jose, CA)
Applicant:	Zoom Communications, Inc. (San Jose, CA)
Family ID:	1000008819326
Assignee:	Zoom Communications, Inc. (San Jose, CA)
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Primary Examiner: Dinh; Khanh Q

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/538,713, filed Nov. 30, 2021, which is a continuation of U.S. patent application Ser. No. 14/577,833, filed Dec. 19, 2014, which claims the benefit of U.S. Provisional Patent Application No. 61/920,698, filed Dec. 24, 2013, the entire disclosures of which are hereby incorporated by reference.

BACKGROUND

(1) The approaches described in this section are approaches that could be pursued, but not necessarily approaches that have been previously conceived or pursued. Therefore, unless otherwise indicated, it should not be assumed that any of the approaches described in this section qualify as prior art merely by virtue of their inclusion in this section.

(2) Web conferencing between multiple participants through a network such as the internet is useful in communicating and sharing ideas even though participants are in remote locations. Some

features of the web conferencing include document sharing, video sharing, screen sharing, and the like.

(3) Participants of web conferencing session can use desktop computers, laptop computers, tablets, smart phones, and smart televisions as their endpoint devices. These endpoint devices can be utilized to display the graphical images from other participants and also utilized to transmit graphical images for display by other participants.

(4) The ease of use of web conferencing encourages participant adoption and allows a participant to share many types of media with other participants.

SUMMARY

(5) The appended claims may serve as a summary of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) In the drawings:

(3) FIG. 1 illustrates a networked computer system with which an embodiment may be implemented.

(4) FIG. 2 illustrates a block diagram according to an embodiment.

(5) FIG. 3 illustrates a block according to an embodiment.

(6) FIG. 4 illustrates a flow diagram according to an embodiment.

(7) FIG. 5 illustrates a computer system with which an embodiment may be implemented.

DETAILED DESCRIPTION

(8) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, that the present invention may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the present invention.

(9) Embodiments are described in sections below according to the following outline:

(10) 1. Overview 2. Direct Integration 3. Gateway 4. Example Embodiments 5. Example Implementation Mechanism—Hardware Overview

1. OVERVIEW

(11) Sharing content within a computer screen is a basic and important feature in an online web conferencing software application. Mobile devices such as tablet PCs and smart phones are widely used in the workplace as well as at home. However, sharing content within the screen of a tablet PC/smart phone may not be available to a web conferencing meeting application running on the mobile device, due to a variety of reasons. For example, mobile devices often have limited operating systems and limited computing resources which make sharing the content with computer screen of the mobile device impractical. Accordingly, online meetings and their associated features may not be available on mobile devices.

(12) A mobile device is usually equipped with a wireless streaming capability that enables streaming or transferring the audio, pictures, videos, and local screen content of the mobile device to another presentation device in real-time or near real-time. Some examples of such streaming protocols implemented in mobile devices include AirPlay, Miracast, Play To, and DLNA.

(13) AirPlay is a streaming protocol defined by Apple to enable an airPlay sender, typically running on an iPod, iPad or iPhone. AirPlay is typically used to stream content such as audio, videos,

photos and screen content from the sender device to an airPlay receiver which typically runs on an Apple TV.

(14) Miracast is a wireless streaming protocol enabling peer-to-peer wireless screencast via Wi-Fi Direct connections. Implemented on Miracast certified desktop PCs, tablets PCs or smart phones, Miracast enables delivery of audio and video from a sending device to a receiving device. Many Android smart phones and Android based smart devices such as projector, smart TV are equipped with Miracast.

(15) Play To is developed by Microsoft and makes it easy to stream music, videos, and photos from a Windows PC to another Windows PC or Windows compatible device such as TV, stereo, game console, tablet, smart phone on a home network.

(16) Originally established by Sony, DLNA defines interoperability guidelines to enable sharing of digital media between multimedia devices such as TV, PC, tablet, smart phone, etc. A DLNA compatible device connects to another in a home network through the UPnP (Universal Plug and Play) protocol. The connection enables playing any media such as audio, video and pictures from the DLNA server (sender) onto the DNL A client (receiver/renderer).

(17) FIG. 1 illustrates a networked computer system with which an embodiment may be implemented. In one approach, a server computer **140** is coupled to a network **130**, which is also coupled to client computers **100**, **110**, **120**. For purposes of illustrating a clear example, FIG. 1 shows a limited number of elements, but in practical embodiments there may be any number of certain elements shown in FIG. 1. For example, the server computer **140** may represent an instance among a large plurality of instances of the application server in a data center, cloud computing environment, or other mass computing environment. There also may include thousands or millions of client computers.

(18) In an embodiment, the server computer **140** hosts a video conferencing meeting and transmits and receives video, image, and audio data to and from each of the client computers **100**, **110**, **120**.

(19) Each of the client computers **100**, **110**, **120** comprises a computing device having a central processing unit (CPU), graphics processing unit (GPU), one or more buses, memory organized as volatile and/or nonvolatile storage, one or more data input devices, I/O interfaces and output devices such as loudspeakers or a LINE-OUT jack and associated drivers. Each of the client computers **100**, **110**, **120** may include an integrated or separate display unit such as a computer screen, TV screen or other display. Client computers **100**, **110**, **120** may comprise any of mobile or stationary computers including desktop computers, laptops, netbooks, ultrabooks, tablet computers, smartphones, etc. Typically the GPU and CPU each manage separate hardware memory spaces. For example, CPU memory may be used primarily for storing program instructions and data associated with application programs, whereas GPU memory may have a high-speed bus connection to the GPU and may be directly mapped to row/column drivers or driver circuits associated with a liquid crystal display (LCD) that serves as the display. In one embodiment, the network **130** is the Internet.

(20) Each of the client computers **100**, **110**, **120** hosts, in an embodiment, an application that allows each of the client computers **100**, **110**, **120** to communicate with the server computer **140**. In an embodiment, the server computer **140** may maintain a plurality of accounts, each associated with one of the client computers **100**, **110**, **120** and/or one or more users of the client computers.

2. DIRECT INTEGRATION

(21) In one embodiment, the streaming receiver is tightly integrated into a web conferencing client for use in a web conferencing meeting application. In this embodiment, a primary device is connected with the web conferencing meeting and a secondary device is associated with the primary device. In one embodiment, the secondary device is a mobile device which is capable of transmitting content in the form of a screen shot to the primary device. In one embodiment, the primary device identifies and seamlessly shares the content from the secondary device with other participants of the web conference. In use, the content from the screen of the secondary device is

transmitted to other participants of the web conference as if the screen content from the secondary device is coming from the primary device. The mobile device screen is considered as one of the applications which, if selected, can be shared out by the meeting client.

(22) FIG. 2 illustrates a block diagram showing exemplary web conferencing architecture. In this embodiment, meeting client devices **220**, **230** and **240** are participants engaged in a web conferencing session through a network **250**. In one embodiment, the screen of a mobile device **210** is shared to meeting client devices **220**, **230** and **240**. The streaming receiver on the meeting client device **220** is built into the meeting client and configured to receive data from a remote device such as the mobile device **210**. In this embodiment, when the screen of the mobile device **210** is streamed to the meeting client device **220**, the web conferencing application running within the meeting client device **220** detects the data which in this case is the screen data of the mobile device **210**. Further, the meeting client device **220** transmits and shares the screen data of the mobile device **210** to meeting client devices **230** and **240** through regular meeting sharing capability within the web conferencing session.

(23) In this embodiment, the mobile device **210** acts as a peripheral device to the meeting client device **220** instead of an independent meeting client. One of the motivations to have the mobile device **210** share through the meeting client device **220** instead of sharing directly with meeting client devices **230** and **240** is that the mobile device A may not have enough processing or memory capacity to operate the web conferencing application to participate directly in the web conferencing session.

3. GATEWAY

(24) In one embodiment, the streaming receiver which is configured to detect and receive data from another device is implemented as a part of a virtual meeting client which is considered a gateway to the on-line meeting. The mobile device screen is considered as the to-be-shared screen of the virtual meeting client.

(25) FIG. 3 illustrates a block diagram with an exemplary architecture in which the screen of mobile device **310** is shared to meeting client devices **320**, **330** and **340** through a meeting gateway **350**. In one embodiment, mobile device **310** and meeting client devices **320**, **330**, and **340** are engaged in a web conferencing session through network **360**. In one embodiment, a virtual meeting client is running on gateway **350** which also acts as a streaming receiver to detect and transmit data streamed from mobile device **310**. When the screen of the mobile device **310** is streamed to gateway **350**, the virtual meeting client running on gateway **350** detects and transmits the data and shares it out to meeting client devices **320**, **330**, and **340** using regular meeting sharing capability.

(26) In this architecture, mobile device **310** acts as an independent meeting client and gateway **350** is not visible to other meeting clients such as client devices **320**, **330**, and **340**. The gateway **350** can be a Windows PC, Mac, iPad, iPhone or Android device, etc.

4. EXAMPLE EMBODIMENTS

(27) FIG. 4, shown below, illustrates a flow diagram that describes various embodiments of a method for streaming content during web a conferencing meeting from a mobile device. Functional blocks are utilized within the flow diagram to demonstrate different aspects; functional blocks may be combined, divided, or removed without narrowing the scope of the embodiments.

(28) In Block **405**, a web conferencing meeting is initiated. In one embodiment, the web conferencing meeting allows participants to share their respective screens and content with other participants. The participants may be in remote locations and communicate through a network such as the internet. In one embodiment, the web conferencing meeting is hosted through the server computer **140** (shown in FIG. 1).

(29) In Block **410**, client devices are detected within the web conferencing meeting. In one embodiment, each of the client devices can be a computer, a tablet, a smartphone, or any network accessible device capable of receiving information and displaying information.

(30) In Block **415**, requests to share data are detected at each client device associated with the web

conferencing meeting. In one embodiment, the web conferencing application runs on each of the client devices.

(31) In Block **420**, if there is a request to share data at a client device, the local content sources are detected for this client device. For example, if the client device is a computer and the client device wishes to share content such as screen sharing, document sharing and the like, the local sources available to this computer are displayed. Local sources associated with this computer can include a webcam coupled to the computer, on-line storage accessible and controlled by the computer, data stored on the computer, mobile devices associated with the computer through DNLA, AirPlay, Miracast or Play To, and other peripheral devices associated with the computer. In another embodiment, local content sources can also include wireless devices located within the proximity of the computer. In this embodiment, an authentication process is utilized to grant permission from the wireless device to be utilized by the computer as a local source.

(32) In one embodiment, the local content sources available to the client device are presented through the client device as potential content source choices. For example, if a smartphone is available as a local content source and photographs are also available stored locally on the computer, then both the smartphone and locally stored photographs are presented as potential content sources for the web conferencing meeting through the client device. In another embodiment, the specific content that is available through the smartphone is individually presented as potential content source choices.

(33) In another embodiment, instead of having the client device being a computer, the client device is a gateway which participates in the web conferencing meeting as a pass through for an associated mobile device. In this embodiment, other meeting participants see the mobile device as the participant even though the gateway is the client device running the web conferencing meeting application.

(34) In Block **425**, one of the local content sources is selected to be shared by the client device with other participants of the web conferencing meeting.

(35) In Block **430**, the content from the selected local content source is streamed through the client device and is available to other participants (other client devices) of the on-line meeting. In one instance, when a mobile device utilized as a local content source for a client device, this mobile device is capable of streaming any of the content that is available on this mobile device. For example, content available on the mobile device that is available for streaming include at least one of and are not limited to: audio content, video content, photographs, text messages, documents, and emails. In another embodiment, video or audio recorded in real-time or near real-time on a mobile device can be streamed through the client device such that the mobile device acts as a wireless webcam or audio recorder for the client device.

5. EXAMPLE IMPLEMENTATION MECHANISM—HARDWARE OVERVIEW

(36) According to one embodiment, the techniques described herein are implemented by one or more special-purpose computing devices. The special-purpose computing devices may be hard-wired to perform the techniques, or may include digital electronic devices such as one or more application-specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs) that are persistently programmed to perform the techniques, or may include one or more general purpose hardware processors programmed to perform the techniques pursuant to program instructions in firmware, memory, other storage, or a combination. Such special-purpose computing devices may also combine custom hard-wired logic, ASICs, or FPGAs with custom programming to accomplish the techniques. The special-purpose computing devices may be desktop computer systems, portable computer systems, handheld devices, networking devices or any other device that incorporates hard-wired and/or program logic to implement the techniques.

(37) For example, FIG. 5 is a block diagram that illustrates a computer system **500** upon which an embodiment of the invention may be implemented. Computer system **500** includes a bus **502** or other communication mechanism for communicating information, and a hardware processor **504**

coupled with bus **502** for processing information. Hardware processor **504** may be, for example, a general purpose microprocessor.

(38) Computer system **500** also includes a main memory **506**, such as a random access memory (RAM) or other dynamic storage device, coupled to bus **502** for storing information and instructions to be executed by processor **504**. Main memory **506** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **504**. Such instructions, when stored in non-transitory storage media accessible to processor **504**, render computer system **500** into a special-purpose machine that is customized to perform the operations specified in the instructions.

(39) Computer system **500** further includes a read only memory (ROM) **508** or other static storage device coupled to bus **502** for storing static information and instructions for processor **504**. A storage device **510**, such as a magnetic disk or optical disk, is provided and coupled to bus **502** for storing information and instructions.

(40) Computer system **500** may be coupled via bus **502** to a display **512**, such as a cathode ray tube (CRT), for displaying information to a computer user. An input device **514**, including alphanumeric and other keys, is coupled to bus **502** for communicating information and command selections to processor **504**. Another type of user input device is cursor control **516**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **504** and for controlling cursor movement on display **512**. This input device typically has two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane.

(41) Computer system **500** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic which in combination with the computer system causes or programs computer system **500** to be a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **500** in response to processor **504** executing one or more sequences of one or more instructions contained in main memory **506**. Such instructions may be read into main memory **506** from another storage medium, such as storage device **510**. Execution of the sequences of instructions contained in main memory **506** causes processor **504** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

(42) The term “storage media” as used herein refers to any non-transitory media (e.g., a non-transitory computer readable medium) that store data and/or instructions that cause a machine to operation in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage device **510**. Volatile media includes dynamic memory, such as main memory **506**. Common forms of storage media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge.

(43) Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **502**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(44) Various forms of media may be involved in carrying one or more sequences of one or more instructions to processor **504** for execution. For example, the instructions may initially be carried on a magnetic disk or solid state drive of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a telephone line using a modem. A modem local to computer system **500** can receive the data on the telephone line and use

an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector can receive the data carried in the infra-red signal and appropriate circuitry can place the data on bus **502**. Bus **502** carries the data to main memory **506**, from which processor **504** retrieves and executes the instructions. The instructions received by main memory **506** may optionally be stored on storage device **510** either before or after execution by processor **504**.

(45) Computer system **500** also includes a communication interface **518** coupled to bus **1002**. Communication interface **518** provides a two-way data communication coupling to a network link **520** that is connected to a local network **522**. For example, communication interface **518** may be an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, communication interface **518** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN. Wireless links may also be implemented. In any such implementation, communication interface **518** sends and receives electrical, electromagnetic or optical signals that carry digital data streams representing various types of information.

(46) Network link **520** typically provides data communication through one or more networks to other data devices. For example, network link **520** may provide a connection through local network **522** to a host computer **524** or to data equipment operated by an Internet Service Provider (ISP) **526**. ISP **526** in turn provides data communication services through the world wide packet data communication network now commonly referred to as the “Internet” **528**. Local network **522** and Internet **528** both use electrical, electromagnetic or optical signals that carry digital data streams. The signals through the various networks and the signals on network link **520** and through communication interface **518**, which carry the digital data to and from computer system **500**, are example forms of transmission media.

(47) Computer system **500** can send messages and receive data, including program code, through the network(s), network link **520** and communication interface **518**. In the Internet example, a server **530** might transmit a requested code for an application program through Internet **528**, ISP **526**, local network **522** and communication interface **518**.

(48) The received code may be executed by processor **1004** as it is received, and/or stored in storage device **510**, or other non-volatile storage for later execution.

(49) In the foregoing specification, embodiments of the invention have been described with reference to numerous specific details that may vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the invention, and what is intended by the applicants to be the scope of the invention, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction.

Claims

1. A method, comprising: identifying, by a primary device, a secondary device as a source of content available for streaming on a web conference; and streaming, by the primary device while the primary device is connected to the web conference and without the secondary device connecting to the web conference, the content on the web conference to one or more other devices connected to the web conference without the secondary device being visible to the one or more other devices.
2. The method of claim 1, wherein identifying the secondary device as the source of the content available for streaming on the web conference comprises: initiating, by the primary device, a detection of secondary devices that are sources of potential content; and receiving, by the primary device based on the detection of the secondary devices, an indication of the secondary device.
3. The method of claim 1, wherein streaming the content on the web conference to the one or more

other devices connected to the web conference comprises: streaming, by the primary device, the content on the web conference via a wireless network between the primary device and the secondary device.

4. The method of claim 1, comprising: indicating, to the one or more other devices, that the content is streamed on the web conference by the primary device.

5. The method of claim 1, comprising: performing an authentication process to enable the primary device to access the content.

6. The method of claim 1, comprising: receiving, by the primary device, a selection of the content from a set of content available at the secondary device.

7. The method of claim 1, wherein the content is screen share content output at a display of the secondary device.

8. The method of claim 1, wherein the content includes at least one of audio content, video content, photographs, text messages, documents, or emails.

9. The method of claim 1, wherein the primary device is a laptop computer or a desktop computer and the secondary device is a tablet computer or a smart phone.

10. A non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising: identifying, by a primary device, a secondary device as a source of content available for streaming on a web conference; and streaming, by the primary device while the primary device is connected to the web conference and without the secondary device connecting to the web conference, the content on the web conference to one or more other devices connected to the web conference without the secondary device being visible to the one or more other devices.

11. The non-transitory computer readable medium of claim 10, wherein identifying the secondary device as the source of the content available for streaming on the web conference comprises: detecting secondary devices based on a share request; and identifying the secondary device from amongst the secondary devices.

12. The non-transitory computer readable medium of claim 10, access to the content of the secondary device at the primary device is enabled based on an authentication process.

13. The non-transitory computer readable medium of claim 10, wherein the content is streamed on the web conference via a wireless network between the primary device and the secondary device.

14. The non-transitory computer readable medium of claim 10, wherein the content is streamed by the primary device using a client application operating as a gateway for the web conference.

15. A system, comprising: a secondary device; and a primary device including a memory and a processor configured to execute instructions stored in the memory to: identify the secondary device as a source of content available for streaming on a web conference; and stream, while the primary device is connected to the web conference and without the secondary device connecting to the web conference, the content on the web conference to one or more other devices connected to the web conference without the secondary device being visible to the one or more other devices.

16. The system of claim 15, wherein the secondary device is identified based on a wireless network via which the primary device and the secondary device are in communication, and wherein the content is streamed on the web conference by the primary device via the wireless network.

17. The system of claim 15, wherein, to identify the secondary device as the source of the content available for streaming on the web conference, the processor is configured to execute the instructions to: present indications of secondary devices including the secondary device for selection at the primary device.

18. The system of claim 15, wherein an indication that the content is streamed on the web conference by the primary device is transmitted to the one or more other devices.

19. The system of claim 15, wherein the primary device is authenticated to access the content of the secondary device.

20. The system of claim 15, wherein the content is selected from a set of content available at the secondary device.
